

# 04: Prediction for SLR

Stat 230 – Spring 2026

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## 1 Warm-up questions

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Is education level associated with income? Researchers collected education level (in years) and income (in thousands of dollars) for a random sample of 32 employees working for the city of Riverview.

The researchers fit a simple linear regression of income on education and obtained the following output:

| term        | estimate | std.error | statistic | p.value |
|-------------|----------|-----------|-----------|---------|
| (Intercept) | 11.321   | 6.123     | 1.849     | 0.074   |
| education   | 2.651    | 0.370     | 7.173     | <0.001  |

**Q1.** Write the fitted regression equation.

**Q2.** Interpret the slope coefficient in the context of the problem. (Don't forget to specify units.)

**Q3.** Interpret the intercept in the context of the problem. (Don't forget to specify units.)

A new researcher joined the team and decided that education should be standardized (to have

mean 0 and SD 1) before fitting the regression model. The output from this regression is shown below:

| term             | estimate | std.error | statistic | p.value |
|------------------|----------|-----------|-----------|---------|
| (Intercept)      | 53.742   | 1.587     | 33.861    | <0.001  |
| scale(education) | 11.567   | 1.613     | 7.173     | <0.001  |

**Q4.** Write the fitted regression equation for this new model.

**Q5.** Interpret the slope coefficient in the context of the problem. (Don't forget to specify units.)

**Q6.** Interpret the intercept in the context of the problem. (Don't forget to specify units.)

**Q7.** Compare the two models. What's the same? What's different?

## 2 Prediction and confidence intervals



Tip

The R code for the following questions is found at  
<https://aluby.github.io/stat230/activities/04R-prediction>  
The URL is also posted on Moodle.

Now, we'll explore the `city` data using R.

**Q1.** Make a plot of `income` against `education` and overlay the linear regression equation. Does it look like what you expected?

**Q2.** Make a plot of `income` against `scale(education)` and overlay the linear regression equation. Does it look like what you expected?

**Q3.** Find the linear regression equation from Q1. Confirm that it matches the table at the beginning of this worksheet.

**Q4.** The first person in this dataset has `education = 8`. Calculate the expected income of this person using the first regression equation.

**Q5.** If we want to predict the income of **this** person, should we use a confidence interval or a prediction interval?

**Q6.** Use R to construct the appropriate 89% interval for this person's income. Record this interval

below.

**Q7.** Interpret the interval in context.

**Q8.** Run the code to produce a scatterplot, regression line, and both types of intervals. Which is the prediction interval and which is the confidence interval? How can you tell?